Printed Pages:02 **Sub Code: NAS -301**

Paper Id: 233323 **Roll No.** | | | | | | | | | | |

B.TECH.
(SEM III) THEORY EXAMINATION 2022-23
MATHEMATICS -III

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

(a) If $u = 3x^2y - y^3$, find the analytic function $f(z) = u + iv$

(b) Find the nature and location of the singularity of $f(z) = \frac{z-sinz}{z^2}$.

(c) State convolution theorem on Fourier transformation.

(d) Define the change of scale property for Z-transform.

(e) Define coefficient of skewness and coefficient of Kurtosis.

(f) Write the formula for rank correlation in tied case.

(g) Write the iteration formula of Newton-Raphson for finding the cube root of N.

(h) Find missing term in the following table

x	0	1	2	3	4
y	1	3	9	-	81

(i) Write Picard Method for first order ordinary differential equation.

(j) Define Simpson's 3/8th rule from Newton-cotes quadrature formula

SECTION B

2. Attempt any *three* of the following: **10x3=30**

(a) Show the function defined by $f(z) = \begin{cases} \frac{x^3y^5(x+iy)}{x^6+y^{10}}, when\ z \neq 0 \\ 0 \qquad\qquad when\ z = 0 \end{cases}$ is not analytic at the origin though it satisfies Cauchy-Riemann equations at the origin.

(b) Find Fourier sin transform of $\frac{e^{-ax}}{x}, a > 0$.Hence find Fourier sine transformation of $\frac{1}{x}$.

(c) using the method of least squares, derive the normal equations to fit the curve $y = ax^2 + bx$.Hence fit this curve to the data:

x:	1	2	3	4	5	6	7	8
y:	1	1.2	1.8	2.5	3.6	4.7	6.6	9.1

(d) Find the real root of the equation$\boldsymbol{xe^x = cos\,x}$by the method of false position correct to four decimal places.

(e) Evaluate$\int_0^6 \frac{e^x dx}{1+x}$by Simpson's 3/8th rule by dividing the range into 6 equal parts.

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SECTION C

3. Attempt any *one* part of the following: 10x1=10

(a) Evaluate the following integral using Cauchy Integral formula $\int_C \frac{z}{(z-1)(z-3)}dz$; where C is the circle$|z| = 4$

(b) Using Complex integration method, evaluate $\int_0^{2\pi} \frac{d\theta}{5-3\cos\theta}$.

4. Attempt any *one* part of the following: 10x1=10

(a) Express the function as a Fourier integral if $F(x) = \begin{cases} 1, & x \leq 1 \\ 0, & x > 1 \end{cases}$.

Hence evaluate $\int_0^{\infty} \frac{\sin\lambda \cos\lambda x \, d\lambda}{\lambda}$

(b) Find the inverse Z-transform of : $F(z) = \frac{1}{(z-3)(z-2)}$for

$|z| < 2$ (ii) $2 < |z| < 3$(iii) $|z| > 3$

5. Attempt any *one* part of the following: 10x1=10

(a) Two random variables have the regression lines with equations and$3x + 12y = 19$ and $9x + 3y = 46$.Find the mean values, correlation coefficient and regression coefficient between x and y

(b) Fit a poisson distribution to the following data and calculate theoretical frequencies

Accidentper shift	0	1	2	3	4
Frequency	192	100	24	3	1

6. Attempt any *one* part of the following: 10x1=10

(a) Using Lagrange interpolation formula find the value of y(5) from the following table

x	1	2	3	4
y	3	1	1	9

(b) Perform 5 iterations of the bisection method to obtain the smallest positive root of the equation $x^3 - 5x + 1 = 0$.

7. Attempt any *one* part of the following: 10x1=10

(a) Use fourth order Runge–Kutta method to find y(0.8) solving $\frac{dy}{dx} = 1 + y^2; y(0) = 0$

(b) Solve the following system of equation by LU Decomposition method:
5x - 2y + 2z = 4 , 7x + y - 5 z = 8 ,3x + 7y + 4 z = 10.

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